



Nato ABC (natoabc)

A text message has been encoded using the NATO phonetic alphabet, where each letter of the original message is replaced by its corresponding NATO code word. During transmission, all spaces between the code words were removed.

Your task is to decode the given continuous string and recover the original text.

The encoding uses the following mappings:

a – alpha	b – bravo	c – charlie	d – delta	e – echo	f – foxtrot
g – golf	h – hotel	i – india	j – juliett	k – kilo	l – lima
m – mike	n – november	o – oscar	p – papa	q – quebec	r – romeo
s – sierra	t – tango	u – uniform	v – victor	w – whiskey	x – xray
y – yankee	z – zulu				

Table 1: NATO Phonetic Alphabet

📎 Among the attachments of this task you may find a template file `natoabc.*` with a sample incomplete implementation.

Input

The input consists of a single line containing the string S , a valid space-free concatenation of NATO phonetic code words using only lowercase English letters.

Output

Output a single line containing the decoded string T of lowercase English letters.

Constraints

- $1 \leq |S| \leq 10\,000$.
- S contains only lowercase English letters.
- S is a valid concatenation of NATO phonetic code words without spaces.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** (0 points) Examples.
- **Subtask 2** (50 points) $N \leq 100$.
- **Subtask 3** (50 points) No additional limitations.

Examples

input	output
kilooscardeltakilouniformpapalpha	kodkupa
indiaindiaoscartango	iiot

Explanation

In the **first sample case**. The input string can be split as: `kilo | oscar | delta | kilo | uniform | papa | alpha` which decodes to: `kodkupa`.

In the **second sample case**. The input string can be split as: `india | india | oscar | tango` which decodes to: `iiot`.